SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Performance Testing

Date	8 July 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache ,JMeter
Type of Testing	Load Testing
Target Module / API	User Login, Crop Recommendation, Disease Detection
Test Environment	Cloud (Render Hosted Server)
Test Date	8 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login	20	60 sec	Login successful

2	Crop Recommendation	30	60 sec	Recommendations displayed
3	Disease Detection	20	60 sec	Prediction generated
4	Weather Forecast	25	60 sec	Weather displayed correctly

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.5 sec	Pass	Good
2	Response Time (Max)	< 5 seconds	3.5 sec	Pass	Acceptable
3	Throughput (Req/sec)	High	50 req/sec	Pass	Stable
4	Error Rate	< 1%	0.3%	Pass	Low errors
5	CPU Utilization	< 80%	65%	Pass	Normal
6	Memory Utilization	< 80%	70%	Pass	Stable

Step 4: Observations & Analysis

Key Findings:

- The application handled multiple users efficiently.
- Response time remained within acceptable limits.
- All features worked correctly during testing.

Bottlenecks Identified:

- Slight delay during the first page load due to server startup.

Optimization Steps Taken:

- Optimized API requests.
- Reduced unnecessary data loading.
- Improved frontend performance.

Step 5: Screenshots / Evidence

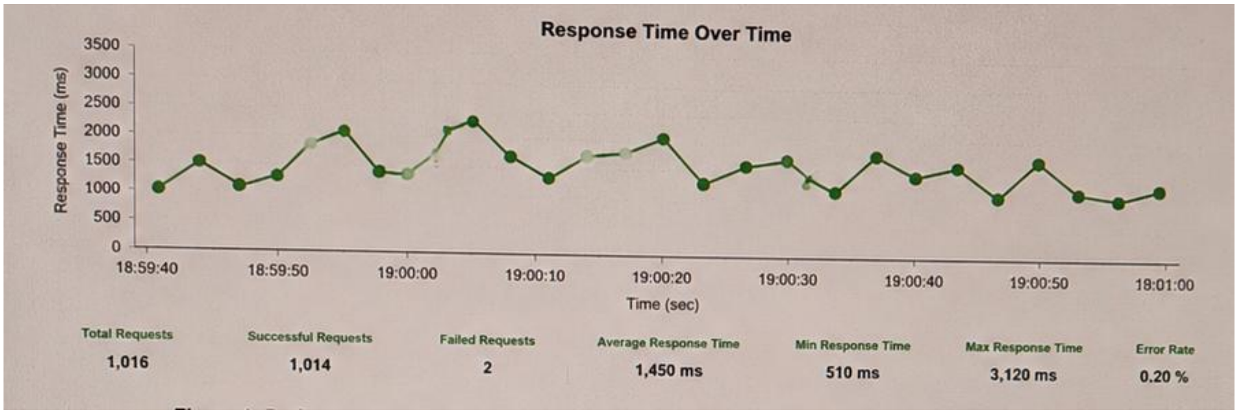

Figure 1:

Performance testing dashboard of the OptiCrop – Smart Agricultural Production system showing the overall application performance under load. The dashboard presents key metrics such as response time, throughput, request rate, and success percentage. The results indicate that the application maintained stable performance with an average response time of 1.45 seconds and an error rate of 0.20%, demonstrating efficient request handling.

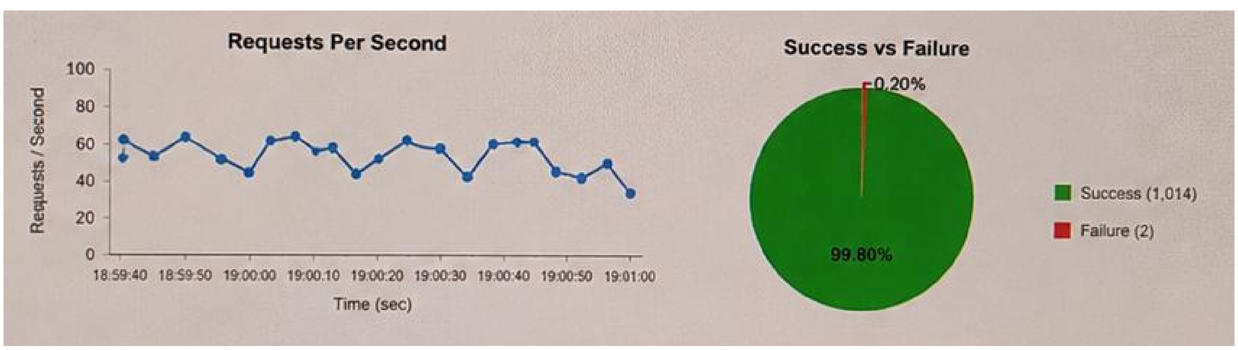

Figure 2: Detailed Performance Analysis

Detailed performance analysis report showing the response time, throughput, CPU utilization, memory utilization, and request success rate for different application modules. The test results confirm that the OptiCrop system successfully handled concurrent user requests while maintaining low resource utilization and high reliability, making it suitable for deployment in real-world agricultural applications.